Task 1 — Normal Equation

1. The Normal Equation

It's an equation used to calculate the coefficients for datasets with small number of features in only one step, making it more efficient in some cases than Gradient decent, which requires many iterations and small steps and therefore more time to do so.

2. Mathematical Derivation and Formula

The normal equation is derived from the idea of solving (the differential formula =0) to get the point where the function is the minimum. The function here is the Mean Squared Error (MSE) function:

$$J(\theta_{0...n}) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2$$ or $$J(\theta) = \frac{1}{2m}(X\theta - y)^T(X\theta - y)$$

, where: $$h_\theta(x) = \theta_0 x_0 + \theta_1 x_1 + \cdots + \theta_n x_n$$

Since the x is presented as a matrix of size (m x (n+1)) and the regression coefficients are in a matrix of size ((n + 1) x 1): $h_\theta(x) = \theta^T x$

Then the MSE function was expanded with transpose identities after discarding the 1/2m factor, since it will be cancelled by equaling the function to 0 later anyways, so we ended up with: $J(\theta) = \theta^T X^T X\theta - 2(X\theta)^T y + y^T y$

Then the function is partially differentiated with respect to theta and equaled to 0:

$$\frac{\partial J}{\partial \theta} = 2X^T X\theta - 2X^T y = 0$$

$$X^T X\theta = X^T y \quad \rightarrow \quad \theta = (X^T X)^{-1} X^T y$$

Where the theta on the left side is a column matrix of size ((n+1) x 1) with all the coefficients of the regression model, looking like this:

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix} \begin{array}{c} \text{(Intercept / Bias)} \\ \text{(Coefficient for Feature 1)} \\ \text{(Coefficient for Feature 2)} \\ \vdots \\ \text{(Coefficient for Feature } n\text{)} \end{array}$$

3. How it’s used in linear regression:

The Normal Equation finds the optimal parameters by using matrix calculus to set the gradient of the cost function to zero, analytically solving for the exact minimum in a single mathematical step.

4. The Normal Equation vs Gradient Decent:

Feature	Gradient Descent	Normal Equation
Hyperparameters	We need to choose the learning rate, Number of iterations, etc.	No hyperparameters are needed.
Approach	It is an iterative algorithm.	It is an analytical approach.
Data Size	It works well with large number of features.	It works well with small number of features.
Feature Scaling	Feature scaling is needed.	No need for feature scaling.
Handling Non-Invertibility	No need to handle non-invertibility cases.	Regularization is needed to handle this.
Time Complexity	Depends upon number of iterations and data size.	Depends upon on the matrix inversion operation of the input feature.

Advantages of The Normal Equation include; being faster, easier to call in the code, and working faster with datasets with small numbers of features. But disadvantages include; consuming way more time with datasets with larger numbers of features, and the fact that it works only for cases where a direct solution exists like in linear regression.

On the other hand, for Gradient Decent, advantages include; being able to function faster than the normal equation when dealing with a large number of features and being able to find the coefficients for cases where no direct answer exists. But disadvantages include; being slower than the normal equation with smaller numbers of features, and it needs tunning before being used in the code.